

however, is used to determine costs for the underlying trading strategy—whether it be a trade schedule, a POV based strategy, or a trading rate based strategy. An order for 100,000 shares or 10% ADV will have different expected costs if the volume on the day is 1,000,000 shares or 2,000,000 shares. In this case, a more accurate daily volume estimate will increase precision in the cost estimate and lead to improved trading performance.

Daily Forecasting Analysis—Methodology

Time Period: Jan. 1, 2011 through Dec. 31, 2011.

Sample Universe: SP500 (large cap) and R2000 (small cap) indexes on Dec. 31, 2011. We only included stocks where we had complete trading history over the period Nov. 10, 2010 through Dec. 31, 2011. The days from Nov. 11, 2010 to Dec. 31, 2010 were used to calculate the starting point for the historical average daily volume (ADV) and historical median daily volume (MDV) on Jan. 1, 2011.

Variable Notation

$V(t)$ = actual volume on day t .

$\hat{V}(t)$ = forecasted volume for day t .

$MDV(n)$ = median daily volume computed using previous n trading days.

$ADV(n)$ = Average daily volume computed using previous n trading days.

$Day\ Of\ Week(t)$ = The percentage of weekly volume that typically trades on the given weekday.

$\hat{\beta}$ = Autoregressive sensitivity parameter—estimated via OLS regression analysis.

$e(t)$ = forecast error on day t .

ARMA Daily Forecasting Model

$$\hat{V}(t) = \bar{V}_t(n) \cdot Day\ Of\ Week(t) + \hat{\beta} \cdot e(t-1) \quad (7.49)$$

where $\bar{V}_t(n)$ is either the n -day moving ADV or n -day moving MDV, and $e(t-1)$ is the previous day's volume forecast error (actual minus estimate). That is

$$e(t-1) = V(t-1) - (\bar{V}_{t-1}(n) \cdot Day\ Of\ Week(t-1)) \quad (7.50)$$

The error term above is calculated as the difference between actual volume on the day and estimated volume only using the day of week